

Transforming Healthcare Performance: AI Analytics as a Driver of Early Diagnosis and Operational Efficiency

A dissertation report submitted in partial fulfilment of the requirements of the degree of Master of Business Administration

Submitted by

Saket Sinha

Registration No.

2412012010019

Under the Guidance of

Dr. Nagaraj S





Master of Business Administration

Declaration

I hereby declare that the dissertation report titled “Transforming Healthcare Performance: AI Analytics as a Driver of Early Diagnosis and Operational Efficiency” submitted in partial fulfilment of the requirements for the award of the degree of Master of Business Administration (MBA) with specialisation in Business Analytics at Alliance University, Alliance Ascent College, is a record of original work carried out by me under the guidance of Dr. Nagaraj S.

I further declare that this report has not been submitted to any other university or institution for the award of any degree or diploma. All sources of information have been duly acknowledged and cited. I have adhered to all principles of academic integrity and have not fabricated or falsified any idea, data, fact, or source in my submission.

Signature of the Student:

Name: Saket Sinha

Registration No.: 2412012010019

Date:



Master of Business Administration

Certificate

This is to certify that Mr. Saket Sinha Regn. No. 2412012010019 has completed the dissertation titled “Transforming Healthcare Performance: AI Analytics as a Driver of Early Diagnosis and Operational Efficiency” under my guidance as part of the dissertation, serving as partial fulfilment of the requirements for the degree of Master of Business Administration.

Signature of the Faculty Guide:

Name of the Faculty Guide:

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Name-Saket Sinha

EXECUTIVE SUMMARY

Healthcare is sitting on a mountain of data — from patient records and diagnostic scans to real-time monitoring streams — and the question is no longer whether that data exists, but whether organisations are doing anything useful with it. This dissertation explores exactly that: how AI-driven analytics is reshaping healthcare, both clinically and as a business.

The research surveyed 150 professionals across ten industries — healthcare, IT, finance, pharma, consulting, and more — asking them to rate their perceptions of AI across four areas: how it's being adopted, what it does for early diagnosis, how it affects operational performance, and what gets in the way.

The headline finding is straightforward: people broadly believe AI works and that it's here to stay. Across all 20 survey questions, the overall agreement score sat at 63.6%, and nearly three-quarters of respondents believe AI adoption in healthcare will grow significantly over the next five years.

Where confidence was highest was in early disease detection. Respondents almost unanimously agreed that catching diseases early — through AI-assisted imaging, pattern recognition, and predictive tools — improves patient outcomes and reduces costs. The two questions with the strongest consensus in the entire study were about the importance of accurate diagnosis (74.7% agreement) and AI's ability to improve medical imaging accuracy (74.7% agreement). The research backs this up: early detection can cut global treatment costs by 30 to 50 percent, and AI systems are already matching the diagnostic accuracy of specialist clinicians in areas like skin cancer, diabetic retinopathy, and sepsis prediction.

On the business side, respondents were equally clear that resource efficiency matters. The statement "resource efficiency is key to business performance" achieved the highest agreement rate in the whole study at 76%, and two-thirds of respondents agreed that AI delivers long-term financial benefits. AI is also seen as a meaningful tool for reducing hospital readmissions (62% agreement) and cutting administrative workload (61%).

But the picture isn't all optimistic. The biggest red flag in the data was workforce readiness. The statement that "healthcare professionals are adequately trained to use AI tools" scored the lowest of all 20 questions — just 3.19 out of 5 — with over a third of respondents actively disagreeing. This is a serious gap, and it applies even among experienced professionals. Having the technology is one thing; having people who know how to use it confidently and correctly is another matter entirely.

Data privacy was the other major concern, with 67% of respondents flagging it as a significant issue, and nearly 43% choosing the strongest possible agreement on that point. This is consistent with the growing weight of global regulation — GDPR, HIPAA, India's DPDP Act — and it signals that organisations can't treat governance as an afterthought.

One of the more interesting findings was that these perceptions didn't vary much by industry, age group, or years of experience. Statistical testing confirmed there were no significant differences between groups. In other words, enthusiasm for AI in healthcare isn't a niche view held by tech-forward younger professionals — it's a broad, cross-generational consensus.

The study draws five practical conclusions. First, AI analytics is now seen as a genuine strategic capability, not just a clinical experiment. Second, early diagnosis is where AI's value is clearest and most accepted. Third, operational gains are real, but they're held back by a training gap that organisations need to take seriously. Fourth, data governance and privacy are non-negotiable foundations for any deployment. And fifth, the long-term direction of travel is clear — almost everyone expects AI adoption to accelerate.

For organisations acting on these findings, the priorities are: build structured AI literacy programmes for staff at all levels, focus initial deployments on high-impact diagnostic use cases, put governance frameworks in place before scaling, adopt phased integration approaches to manage technical complexity, and commit to a multi-year analytics maturity roadmap rather than hoping for quick wins.

The global AI healthcare market is expected to grow from around USD 11 billion to USD 187 billion by 2030. The organisations that invest now in people, governance, and the right applications will be the ones that turn that investment into lasting clinical and business performance.

TABLE OF CONTENTS

Chapter 1: Introduction	09
1.1 Background and Context	10
1.2 Problem Statement	10
1.3 Research Motivation	10
1.4 Research Objectives	11
1.5 Scope of the Study	11
1.6 Significance of the Study	12
Chapter 2: Review of Literature	13
2.1 AI and Business Analytics in Healthcare	13
2.2 AI in Early Disease Detection	14
2.3 AI in Operational Efficiency and Business Performance	14
2.4 Theoretical Frameworks	15
2.5 Research Gap	16
Chapter 3: Research Methodology	17
3.1 Research Design	17
3.2 Data Collection	17
3.3 Sampling Design	17
3.4 Research Instrument	18
3.5 Variables Measured	18
3.6 Data Analysis Techniques	18
3.7 Tools Used — Excel, Python and Dashboard	19
Chapter 4: Data Analysis and Interpretation	21
4.1 Demographic Profile of Respondents	21
4.2 Descriptive Statistics — All 20 Questions	22
4.3 Frequency Distribution Analysis	24
4.4 Construct-Level Analysis	27
4.5 Reliability Analysis — Cronbach's Alpha	28
4.6 Demographic Cross-Tabulation Analysis	29
4.7 Hypothesis Testing — ANOVA and Kruskal-Wallis	31
4.8 Correlation Analysis	32
4.9 Dashboard Analysis and Visualisation	34
4.10 Summary of Key Findings	36
Chapter 5: Conclusions and Recommendations	38
5.1 Conclusions	38
5.2 Strategic Recommendations	39
5.3 Limitations of the Study	41
5.4 Scope for Future Research	42

References.....	43-44
Appendix A: Survey Questionnaire.....	45-51
Appendix B: Python Analysis Code Summary	52-57

CHAPTER 1: INTRODUCTION

The intersection of Artificial Intelligence (AI), big data and advanced analytics is revolutionising the world of healthcare. In an era where hospitals and other health care institutions produce vast amounts of structured and unstructured data, ranging from electronic health records (EHRs) and diagnostic images to genomics data and streams of real-time patient monitoring, the real question is no longer the availability of data but how to obtain information. Business analytics, defined as an iterative and systematic process for exploration of organisational data to support decision-making, is a critical strategic skill in healthcare today.

The global market for AI in healthcare will be worth about USD 11 billion in 2021 and is expected to reach USD 187 billion by 2030 with a compound annual growth rate (CAGR) of more than 37% (Grand View Research, 2022). This rapid expansion is a clear sign of a paradigm shift in the way healthcare organisations tackle the challenges of both clinical and operational management - from gut feel to algorithm-driven decision-making. Central to this shift are predictive analytics and machine learning (ML) algorithms that can identify patterns of disease, predict patient risk and allocate resources with a level of accuracy never before achieved.

Early disease detection is one of the most promising use-cases for AI in healthcare. Diseases like cancer, diabetes, heart disease and sepsis pose a greater risk of death when they are detected late; early detection has proven to result in better patient outcomes and lower costs for treatment. Thanks to AI systems trained on large-scale clinical data, it's now possible to detect these diseases prior to the manifestation of symptoms, potentially leading to better outcomes and reduced health-care costs. Early detection of non-communicable diseases is estimated to save 30-50% of the global burden of treatment costs, according to the World Health Organization (WHO, 2020).

In addition to patient benefits, AI analytics is transforming the business landscape of health organisations. Analytics solutions deployed within health systems now support predictive scheduling, smart patient throughput, fraud prevention, supply chain management and real-time financial forecasting. Adoption of these technologies has been shown to improve key business performance measures such as lower readmissions and average length of stay, and more efficient resource allocation.

This research sits at the heart of this shift by considering AI-powered analytics not just as a clinical aid, but as a business analytics strategic capability with a demonstrable impact on organisational performance. As a Business Analytics researcher, the aim is to employ quantitative analytical techniques such as statistical analysis using Python and interactive data visualisation to empirically investigate stakeholder perceptions of the impact of AI across five areas of performance: adoption, early detection, operational efficiency, business performance and governance.

1.1 Problem Statement

While the use of AI-enabled analytics tools is increasingly common in the healthcare industry, there remains a key gap in the empirical body of knowledge: the majority of current research assesses the impact of AI in healthcare from a purely technical or clinical perspective, based on the accuracy of algorithms, model architecture and sensitivity of diagnosis. There is a lack of research that explores the simultaneous impact of AI-powered early diagnosis on business performance measures (including cost per patient, readmission rate and throughput) from an integrated perspective. Moreover, there is a lack of research on the link between analytics maturity and business performance in healthcare organisations, especially in mid-market and emerging-market health care models.

This research seeks to bridge this gap by exploring AI-enabled analytics from a holistic business performance perspective rooted in business analytics, through structured primary research and quantitative analysis.

1.2 Research Motivation

The impetus for this research comes from two sources. First, the growth and adoption of AI in healthcare is unparalleled but the business value of AI investment is not well documented in academic research. Second, as a Business Analytics student in the MBA program at Alliance University, there is a personal academic interest in using analytical methodology - such as Python-based data analytics, reliability testing, and hypothesis testing - to address organisational and strategic challenges in high-impact industries.

Healthcare is a globally significant, data-rich, but little-understood industry. An opportunity to explore the role of data-driven decision-making methodologies in harnessing the power of AI to

deliver tangible value to organisations is of academic and practical interest to healthcare managers, policymakers and technologists alike.

1.3 Research Objectives

The research focus of this study is on the following five objectives:

1. To identify the AI tools and techniques used in healthcare organisations for early diagnosis.
2. To explore the impact of organisational characteristics (data quality, IT infrastructure and analytics capabilities) on AI analytics adoption.
3. To analyse the effect of AI-driven early diagnosis on business performance metrics such as cost savings, readmission rates and efficiency.
4. To understand the association between predictive analytics and increased patient throughput and resource optimisation.
5. To offer insights for health-care organisations that wish to use AI analytics to enhance both clinical and business metrics.

1.4 Scope of the Study

The aim of this study is to understand the attitude and experience of industry professionals from ten different industries - Healthcare, IT/Technology, Finance, Pharmaceuticals, Education, Consulting, Manufacturing, Insurance, Retail and Government - who have been exposed to AI-based analytics for health-related purposes. The study uses primary survey data from 150 participants, using a questionnaire with 20 Likert scale items.

The study has a wide geographic reach, including professionals from various industries and not limited to a particular location or healthcare system. The time frame is 2024-2025, capturing perceptions during a critical period in the adoption of AI. The research does not include any proprietary or confidential data from a healthcare organisation.

1.5 Significance of the Study

This research adds several significant contributions to the domains of business analytics and health care management:

- It connects clinical AI applications with business performance evaluation by adopting the Balanced Scorecard and DDDM approaches to AI-driven healthcare analytics results.
- It offers empirically validated insights (using Cronbach's Alpha, ANOVA and cross-tabulation analysis) into the perceived success of AI tools in terms of key performance metrics.
- It showcases the use of Python for conducting business analytics studies involving frequency analysis, descriptive statistics, reliability analysis, and hypothesis testing using primary data.
- It provides strategic insights for health administrators, policy makers and analytics professionals in a variety of organisational settings.
- It adds to the emerging body of MBA research combining quantitative business analytics with real healthcare management practices.

CHAPTER 2: REVIEW OF LITERATURE

2.1 AI and Business Analytics in Healthcare

The emerging AI and data analytics technologies are transforming the healthcare landscape. The growing adoption of electronic health records, medical imaging systems, wearables and hospital management systems generate huge volumes of data. However, conventional methods of data analytics are inadequate to elicit findings on this abundance of data, leading to late-diagnosis, excessive operations, and rising expenses. AI-related analytics is able to circumvent this issue by assisting organisations in managing large volumes of data, detecting disease patterns in their initial phases and extracting insights that can be used directly in business decision making. In terms of business analytics, the AI applications in health care Business Intelligence generally operate at the three levels of analytics: descriptive analytics, which provides the summarisation of the performance in the past; predictive analytics that forecasts the risk of patients and the resources and prescriptive analytics that suggests the most optimal clinical and operational course of action. Research is constantly confirming that those companies that embrace these abilities achieve positive results in the cost cutting, re-admission rate, average length of stay and resource use (Davenport and Harris, 2007). The use of AI in healthcare has also been democratised, with the introduction of a cloud analytics software, natural language processing, and edge computing. Smaller and mid-tier medical organisations that in the past lacked the data processing advantage of academic medical centres have access to pre-trained instances of AI via API-based services, thus lowering the technical barrier to using AI. This tendency is reflected in the emergence of more types of sectors in the literature on AI in healthcare, which has been revealed in this paper.

2.2 AI in Early Disease Detection

The early detection of acute and chronic disease is probably the most famous and the most noticeable use of AI in health care. Deep learning algorithms, machine learning systems and natural language processing systems have shown themselves to be effective in diagnosing disease conditions like cancer, diabetes, heart disease and sepsis at least as well as clinical physicians. Esteva et al. (2017) demonstrated that a deep learning system that has been trained using over 129,000 clinical images could diagnose skin cancer with the same accuracy as board-certified dermatologists. As shown by Gulshan et al. (2016), AI systems can effectively diagnose diabetes retinopathy using retinal images with high sensitivity and specificity. Obermeyer and Emanuel (2016) established that ML models worked better than traditional regression models in

predicting the probability of sepsis and acute kidney injury 24 hours before their diagnosis. All these studies substantiate the opinion that AI-based early diagnosis is not only a medical breakthrough but also helps directly to business results because interventions in the early stages are 40 to 60 percent cheaper than those in the late stage of diagnosis (American Cancer Society, 2019). In medical imaging, including radiology, histopathology and ophthalmology, the use of AI has had a particularly considerable effect. According to the survey conducted by Rajpura (2022), 130 hospitals with integrated AI diagnostic analytics showed a 18 percent decrease in the percentage of diagnostic errors and a 22 percent decrease in time to diagnosis. It is significant to the performance of the business, as the lower the rates of diagnostic errors, the less the risk of liability, legal expenses and re-hospitalisation expenses, and the faster the diagnosis the higher the number of patients the business can handle and the higher the bed occupancy rates.

2.3 AI in Operational Efficiency and Business Performance

Besides health diagnostics, AI analytics are utilized in service efficiencies and patient throughout, Mesko (2018) reports that predictive algorithms that optimise staff scheduling helped reduce hospital labour costs by an average of 10 to 15 percent, and patient throughout by half a day per year, enabled by AI-assisted discharge planning can release thousands of bed days a year in a single hospital .

Bates (2014) found 10 priority areas in which big data analytics could apply to healthcare, with early risk identification and predictive care management taking the first place in the priority list. The authors observed that hospitals could gain significant efficiency and productivity gains, up to 20 percent (and important cost savings) of avoidable admissions using predictive analytics to identify high-risk patient's infield the framework of the current study.

More recently, the economic ROI of AI in healthcare has been defined early with studies indicating that between US dollars generated between US\$3 and US\$5 of every dollar spent on an AI diagnostic tool and future growth.

2.4 Theoretical Frameworks

This study is based on three theoretical frameworks, which in combination offer a comprehensive perspective in analysing AI analytics as a strategic driver of business performance.

Data-Driven Decision-Making (DDDM) Framework (Brynjolfsson and McElheran, 2016) is based on the assumption that organisations that actively integrate data into decision-making activities deliver high operational and financial performance. Brynjolfsson and McElheran, in a groundbreaking study of more than 4,000 US companies, discovered that the adoption of DDDM was linked to a 5 to 6 percent productivity boost, regardless of the level of IT investments in the company. Within the healthcare setting, the assumption that AI-generated diagnostic and operational information can substitute or supplement the intuition-based clinical and managerial decision-making and result in a statistically significant improvement is supported by DDDM.

The Resource-Based View of the firm (Barney, 1991) suggests that a firm can create sustained competitive advantage based on rare, valuable, inimitable, and non-substitutable (VRIN) organisational resources. When applied to healthcare, AI analytics capability can be viewed as a strategic resource, which meets all four VRIN criteria: it is rare (not all providers implemented advanced analytics), valuable (it enhances clinical and financial performance), inimitable (proprietary training data and institutional workflows make it hard to replicate), and non-substitutable (there is no alternative resource category with the same combination of diagnostic accuracy and operational optimisation). This model supports the thesis according to which AI analytics investment is a true strategic differentiator, but not an operational instrument.

The Balanced Scorecard (BSC) model (Kaplan and Norton, 1996) is a multi-dimensional performance measurement model in four perspectives: financial perspective (cost reduction, ROI), customer perspective (patient outcomes, satisfaction), internal processes perspective (operational efficiency, diagnostic accuracy), and learning and growth perspective (staff training, analytics maturity). This research paper relates to the BSC framework to make sure that AI analytics is considered not only based on financial indicators but on a set of comprehensive organisational performance aspects. The four constructs of this paper Technology and Adoption, Early Diagnosis, Business Performance, and Challenges are directly aligned with the four BSC perspectives.

Davenport and Harris (2007) came up with a five stage analytics maturity model that illustrates the evolution of organisations out of descriptive analytics (what happened) to predictive analytics (what will happen) to prescriptive analytics (what should we do?). The authors noted that most healthcare organisations are yet to reach the first and second phases, which restricts their ability to implement advanced AI solutions. Adler-Milstein. The results presented by

(2017) reviewed fragmented IT systems as the main obstacle to AI scaling in healthcare and the lack of interoperability, which are findings that directly apply to the challenge of governance and integration studied in this paper.

2.5 Research Gap

Despite the existence of substantial research on AI in healthcare, the majority of studies analyse AI tools in purely technical or clinical terms in terms of model accuracy, algorithm design, and diagnostic sensitivity, without creating linkages between AI application and organisational business performance outcomes. Empirical studies that analyze the effects of AI-based early diagnosis on business metrics critical to business operations — cost per patient, readmission rates, and operational throughput — in a single analytical framework are limited.

In addition, interaction between analytics maturity and business performance within healthcare organisations is under-researched, especially within a healthcare environment of mid-size and emerging markets. The bulk of current quantitative research has been restricted to either certain types of disease (oncology, cardiology), or individual case studies by single institutions, which minimise generalisability. Lack of studies utilizing multi-construct reliability testing and hypothesis testing to support internal consistency and cross-demographic robustness of AI perception surveys in healthcare also exists.

The current research fills these gaps by considering AI-informed analytics through the integrated business performance perspective, which involves Cronbach Alpha to validate the reliability, ANOVA to test the cross-group significance, and analysis at the construct level based on the Balanced Scorecard framework.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design

This research design is descriptive and quantitative. The descriptive design would be suitable based on the purpose of the study which is to describe the perceptions of the respondents towards AI-driven analytics in healthcare and its effect on business performance in a systematic manner. The quantitative method allows analysis of structured survey data on a statistical basis in order to determine the patterns, central tendencies, and distributions of responses to specific variables.

The cross-sectional design is used, in which data is gathered on all respondents at one time. This fits well to provide a picture of industry perceptions at a particular stage of AI adoption in healthcare (2024/2025). This approach is based on the positivist research philosophy, which perceptions are measurable variables and can be quantified, analysed and generalised within the limits of sample.

3.2 Data Collection

Structured questionnaire was used to gather primary data by using Google Forms. The survey link was shared via professional networks, LinkedIn, and academic circles that included people with working experience of or exposure to AI and analytics in a healthcare or technology setting. The questionnaire lasted eight weeks and all answers were anonymous and free.

Peer reviewed journal articles, textbooks, industry reports, and online academic databases such as Google Scholar, PubMed, IEEE Xplore, Scopus and Emerald Insight were used as sources of secondary data. The theoretical framework, research gap, and contextualisation of the primary findings were developed using secondary sources.

3.3 Sampling Design

The purposive sampling strategy was utilized, as the professionals who were aware of or exposed to AI analytics in healthcare, technology, or similar areas were targeted. The sample used was 150 respondents and these were in ten industries; Healthcare, IT/Technology, Finance, Pharmaceuticals, Education, Consulting, Manufacturing, Insurance, Retail, and Government.

The sample of 150 respondents is supported by the accepted norms of MBA dissertation research and has sufficient statistical power in descriptive and reliability analysis and inferential analysis (Malhotra and Dash, 2016). The sample will cover diverse age, roles (Managers, Analysts,

Executives, Doctors, IT Specialists, Consultants, Nurses, Administrators) and experience (under 1 year to over 10 years).

3.4 Research Instrument

The main data collection tool will be a structured questionnaire in two parts. Section A touches upon demographic variables: gender, age category, industry, job position, and experience. B section has 20 Likert scale items with a scale of 5 that includes Strongly Disagree (1) and Strongly Agree (5). The questions will be categorized into four constructs that are compatible with Balanced Scorecard framework: Technology and Adoption (Q1, Q2, Q7, Q8, Q11, Q12), Early Diagnosis (Q3, Q4, Q5, Q6, Q19), Business Performance (Q9, Q10, Q14, Q17, Q18, Q20), and Challenges.

3.5 Variables Measured

The following key variables are examined in this study:

- AI-Driven Analytics Adoption in Healthcare (Q1, Q11): Scale of perceived level of AI analytics adoption and predictive analytics implementation in healthcare.
- Effectiveness of Technological Tools (Q2, Q7, Q8, Q12): Measures the perceptions of technological tools contributing to the ease of operation, decision-making processes, and patient safety.
- Early Disease Identification and Diagnostic Outcomes (Q3, Q4, Q5, Q6, Q19): The study explores the beliefs regarding the importance of early detection to improve patient outcomes, reduce costs, and improve diagnostic accuracy.
- Cost Reduction and Operational Efficiency (Q9, Q17, Q18): Perceptions of AI impact on resource use, administrative workload, and financial results, in the long term.
- Difficulties in AI Adoption (Q13, Q15, Q16): Measures perceived obstacles such as sufficient training, complexity of system integration, and concerns about the security of the data.
- General Business Performance/Future Outlook (Q10, Q14, Q20): Remarks the perceived value of AI analytics to the general business performance and future adoption prospects.

3.6 Data Analysis Techniques

The following analytical techniques were applied to the primary data:

- **Frequency Distribution and Percentage Analysis:** The number and percentage of each Likert response (Strongly Disagree to Strongly Agree) to every 20 questions, the building block of descriptive layer of Chapter 4.
- **Descriptive Statistics:** Mean, standard deviation, variance, median, and mode calculated per question and each construct, which allows one to compare the central tendency and dispersion of variables.
- **Construct-Level Composite Scoring:** It consisted of items summarized into four theoretical constructs; composite mean scores calculated per respondent and combined to obtain construct-level means and agreement rates.
- **Reliability Analysis - Cronbachs Alpha:** Internal consistency of each construct was assessed through the use of Cronbachs Alpha coefficient; this coefficient 0.7 and above shows acceptable reliability of Likert scale instruments.
- **Demographic Cross-Tabulation:** Construct means value in comparison to the industry, level of experience, and age group with grouped descriptive statistics and grouped bar/line charts.

Hypothesis Testing: One-WAY ANOVA and Kruskal-Wallis: Inferential tests will be used to test the hypothesis of whether there are significant differences in construct scores by groups of demographics; significant level at $p = 0.05$.

- **Pearson Correlation Analysis:** Pairwise correlation coefficients calculated with all 20 questions to test the inter-variable relationship.

3.7 Tools Used — Excel, Python and Power BI (for Dashboard)

This study was quantitatively analyzed in Microsoft Excel, Python (Jupyter Notebook environment) and Power BI. Initial data cleaning, response coding, preliminary tabulation, and survey dataset preparation were done with Excel. Advanced statistical analysis, reliability testing, correlation analysis, hypothesis testing, and data visualisation were done in Python. The

Power BI was used to create an interactive analytics dashboard to display insights and graphical interpretations of the results.

- Pandas: Loading, cleaning, grouping and tabulating survey results.
- Numpy: Matrix and numerical operations and computations.
- Matplotlib and seaborn: Bar charts, line charts, grouped visualisations, and heatmaps are generated.
- Scipy.stats: Kruskal-Wallis (kruskal) and one-way ANOVA hypothesis testing.
- Pingouin: Cronbach Alpha calculation of construct level reliability analysis.

Moreover, a multi-tab interactive analytics dashboard was created in Business Intelligence software (Power BI). There are five views on the dashboard: Overview (KPI tiles, mean score chart, agreement rates), Question Analysis (individual question distributions, stacked Likert bars), Demographics (gender, age, industry, experience breakdowns), Constructs (radar chart, construct bar chart, ranked table), and Key Findings (written findings with supporting data). As Figures 1 to 12 in Section 4.9, dashboard screenshots are provided.

CHAPTER 4: DATA ANALYSIS AND INTERPRETATION

4.1 Demographic Profile of Respondents

The study collected responses from 150 participants across varied professional backgrounds. The demographic distribution is presented in Table 1.

Table 1: Demographic Profile of Respondents

Category	Sub-Category	Frequency (n)	Percentage (%)
Gender	Male	33	22.0
	Female	62	41.3
	Prefer not to say	55	36.7
Age Group	18–25 years	29	19.3
	26–35 years	32	21.3
	36–45 years	31	20.7
	46–55 years	29	19.3
	56+ years	29	19.3
Industry	IT/Technology	18	12.0
	Retail	18	12.0
	Manufacturing	17	11.3
	Education	17	11.3
	Healthcare	16	10.7
	Finance	15	10.0
	Consulting	15	10.0
	Pharma	14	9.3
	Government	11	7.3
	Insurance	9	6.0
Experience	< 1 year	30	20.0
	1–3 years	36	24.0
	3–5 years	25	16.7
	5–10 years	31	20.7
	10+ years	28	18.7

The sample is remarkably well-balanced across all demographic dimensions. The age groups are nearly uniformly distributed with no single group exceeding 21.3%, ensuring that no age cohort

dominates the findings. The distribution across ten distinct industries, with no single industry exceeding 12%, is a notable strength of the study — it ensures that the findings are not biased towards healthcare professionals alone but reflect a genuinely cross-sectoral perspective on AI analytics. Experience levels are similarly distributed, with the 1–3 years bracket being the most represented (24%), followed by 5–10 years (20.7%) and under 1 year (20.0%).

The gender distribution is notable: female respondents (41.3%) outnumber male respondents (22.0%), with a substantial proportion (36.7%) preferring not to disclose gender. This reflects the evolving nature of professional survey participation and ensures that gender-based analysis incorporates a genuinely diverse set of perspectives. See Figure 8 in Section 4.9 for the demographic visualisations generated through the analytics dashboard.

4.2 Descriptive Statistics — All 20 Questions

Table 2 presents full descriptive statistics for all 20 survey questions. The mean score represents the average Likert response on a 1–5 scale. The standard deviation (SD) measures the spread of responses — a higher SD indicates greater disagreement among respondents. All statistics were computed using Python’s pandas library.

Table 2: Descriptive Statistics — All 20 Survey Questions

Q	Question (Abbreviated)	Mean	SD	Var	Median	Mode	Agree %
Q1	AI adoption in healthcare organisations	3.54	1.09	1.19	4	4	58.7%
Q2	Tech tools improve efficiency	3.85	1.03	1.06	4	4	70.0%
Q3	Early disease ID improves outcomes	3.92	1.10	1.21	4	5	74.0%
Q4	Analytical tools aid early detection	3.41	1.12	1.25	3	4	54.0%
Q5	Early diagnosis reduces cost	3.72	1.08	1.17	4	4	67.3%
Q6	Accurate diagnosis improves quality	3.99	0.95	0.90	4	5	72.7%
Q7	Advanced tech improves efficiency	3.53	1.04	1.08	4	4	58.7%
Q8	Tech insights aid decision-making	3.43	1.16	1.35	3	3	54.0%
Q9	Resource efficiency key to performance	3.90	1.02	1.04	4	4	76.0%

Q	Question (Abbreviated)	Mean	SD	Var	Median	Mode	Agree %
Q10	Analytics improves org performance	3.49	1.19	1.42	4	4	57.3%
Q11	AI reduces readmission rates	3.70	0.97	0.94	4	4	62.0%
Q12	Real-time AI improves patient safety	3.80	1.11	1.23	4	4	71.3%
Q13	Professionals adequately trained	3.19	1.04	1.08	3	2	45.3%
Q14	AI personalises treatment plans	3.51	1.10	1.21	4	4	61.3%
Q15	AI integration is complex	3.48	1.18	1.39	4	4	59.3%
Q16	Data privacy is a major concern	3.89	1.07	1.14	4	5	67.3%
Q17	AI reduces admin burden	3.53	1.12	1.25	4	4	61.3%
Q18	AI gives long-term financial benefits	3.67	1.06	1.12	4	4	66.7%
Q19	AI improves medical imaging accuracy	3.95	0.88	0.77	4	4	74.7%
Q20	AI adoption will grow in 5 years	3.95	1.11	1.23	4	5	74.7%

Table 3: Standard Deviation Interpretation Guide

SD Range	Interpretation	Questions in This Range
SD < 0.95	High consensus, low disagreement	Q6 (0.95), Q11 (0.97), Q19 (0.88)
SD 0.95–1.10	Moderate consensus	Q1, Q2, Q3, Q5, Q7, Q9, Q16, Q18
SD > 1.10	Higher divergence of opinion	Q4, Q8, Q10, Q12, Q14, Q15, Q17, Q20

The descriptive statistics reveal that questions related to the foundational importance of accurate diagnosis (Q6, SD = 0.95) and AI in medical imaging (Q19, SD = 0.88) show the lowest standard deviations, indicating the highest levels of consensus. Conversely, questions relating to decision-making support (Q8, SD = 1.16), system integration complexity (Q15, SD = 1.18), and overall performance impact (Q10, SD = 1.19) show higher dispersion, suggesting more polarised views.

The overall mean across all 20 questions is 3.67, indicating a moderately positive overall perception of AI analytics in healthcare.

4.3 Frequency Distribution Analysis

This section presents the detailed frequency distribution for all 20 survey questions, organised by study construct. For each question, the count and percentage of respondents selecting each Likert option (1 = Strongly Disagree to 5 = Strongly Agree) are reported. These distributions form the primary empirical basis for Chapter 4 and directly address Research Objectives 1, 3, and 4.

4.3.1 Technology and Adoption Construct (Q1, Q2, Q7, Q8, Q11, Q12)

Table 4: Frequency Distribution — Technology & Adoption Construct

Q	Question (Abbreviated)	SD n(%)	D n(%)	N n(%)	A n(%)	SA n(%)	Mean (SD)
Q1	AI adoption in healthcare	9 (6.0)	18 (12.0)	35 (23.3)	59 (39.3)	29 (19.3)	3.54 (1.09)
Q2	Tech tools improve efficiency	5 (3.3)	16 (10.7)	24 (16.0)	56 (37.3)	49 (32.7)	3.85 (1.03)
Q7	Advanced tech improves eff.	8 (5.3)	19 (12.7)	35 (23.3)	62 (41.3)	26 (17.3)	3.53 (1.04)
Q8	Tech insights aid decisions	16 (10.7)	16 (10.7)	37 (24.7)	49 (32.7)	32 (21.3)	3.43 (1.16)
Q11	AI reduces readmission rates	4 (2.7)	16 (10.7)	37 (24.7)	57 (38.0)	36 (24.0)	3.70 (0.97)
Q12	Real-time AI improves safety	11 (7.3)	13 (8.7)	19 (12.7)	59 (39.3)	48 (32.0)	3.80 (1.11)

The Technology and Adoption construct reveals a generally positive but nuanced response pattern. Q2 (Tech tools improve efficiency) achieves the highest agreement rate in the construct at 70.0%, while Q8 (Tech insights aid decision-making) shows the most divergence with a combined disagree rate of 21.4% and the highest SD (1.16) in the construct. This suggests that while respondents broadly accept technology as an efficiency enabler, its role in directly supporting nuanced decision-making is more contested — a finding consistent with the literature noting that AI-generated recommendations are not always trusted or acted upon by clinical professionals (Topol, 2019).

Q11 (AI reduces readmission rates) and Q12 (Real-time monitoring improves safety) both record over 62% agreement, affirming that predictive analytics and real-time data monitoring are seen as valuable tools for patient safety and outcome improvement. The construct mean of 3.64 indicates moderate-to-strong positive consensus.

4.3.2 Early Diagnosis Construct (Q3, Q4, Q5, Q6, Q19)

Table 5: Frequency Distribution — Early Diagnosis Construct

Q	Question (Abbreviated)	SD n(%)	D n(%)	N n(%)	A n(%)	SA n(%)	Mean (SD)
Q3	Early disease ID improves outcomes	8 (5.3)	16 (10.7)	15 (10.0)	52 (34.7)	59 (39.3)	3.92 (1.10)
Q4	Analytical tools aid detection	14 (9.3)	22 (14.7)	33 (22.0)	51 (34.0)	30 (20.0)	3.41 (1.12)
Q5	Early diagnosis reduces cost	9 (6.0)	19 (12.7)	21 (14.0)	57 (38.0)	44 (29.3)	3.72 (1.08)
Q6	Accurate diagnosis improves quality	3 (2.0)	12 (8.0)	26 (17.3)	51 (34.0)	58 (38.7)	3.99 (0.95)
Q19	AI improves medical imaging acc.	1 (0.7)	13 (8.7)	24 (16.0)	66 (44.0)	46 (30.7)	3.95 (0.88)

The Early Diagnosis construct produces the highest agreement rates in the entire study. Q6 (Accurate diagnosis is essential for improving healthcare quality) achieves a mean of 3.99 with only 10.0% disagreement — the lowest disagree rate across all 20 questions. Q19 (AI improves medical imaging accuracy) similarly shows exceptional consensus, with 74.7% agreement and an SD of just 0.88, the lowest in the study. These findings reflect near-universal recognition that diagnostic quality is the foundation of healthcare performance and value creation.

Q3 (Early identification of diseases improves patient outcomes) records the highest proportion of Strongly Agree responses (39.3%), indicating strong conviction rather than merely moderate agreement. Q4 (Analytical tools help detect diseases at an earlier stage) is the only question in this construct scoring below 3.5, with 24% disagreement, reflecting awareness that AI diagnostic deployment is not yet uniform. Q5 (Early diagnosis reduces cost) achieves 67.3% agreement, providing empirical support for the WHO’s projection that early detection could reduce global treatment costs by 30–50%.

4.3.3 Business Performance Construct (Q9, Q10, Q14, Q17, Q18, Q20)

Table 6: Frequency Distribution — Business Performance Construct

Q	Question (Abbreviated)	SD n(%)	D n(%)	N n(%)	A n(%)	SA n(%)	Mean (SD)
Q9	Resource efficiency key to performance	8 (5.3)	13 (8.7)	15 (10.0)	64 (42.7)	50 (33.3)	3.90 (1.02)
Q10	Analytics improves org performance	15(10.0)	23 (15.3)	26 (17.3)	45 (30.0)	41 (27.3)	3.49 (1.19)
Q14	AI personalises treatment plans	11 (7.3)	25 (16.7)	22 (14.7)	60 (40.0)	32 (21.3)	3.51 (1.10)
Q17	AI reduces admin burden	15(10.0)	14 (9.3)	29 (19.3)	61 (40.7)	31 (20.7)	3.53 (1.12)
Q18	AI gives long-term financial benefits	8 (5.3)	22 (14.7)	20 (13.3)	61 (40.7)	39 (26.0)	3.67 (1.06)
Q20	AI adoption will grow in 5 years	12 (8.0)	10 (6.7)	16 (10.7)	48 (32.0)	64 (42.7)	3.95 (1.11)

The Business Performance construct shows a bifurcated pattern. Q9 (Resource efficiency is important for business performance) achieves the highest agreement rate in the entire study at 76.0%, with 75% of respondents choosing Agree or Strongly Agree and only 14% disagreeing. This near-consensus finding affirms that resource optimisation — a core output of AI analytics — is universally valued as a business performance lever.

Q20 (AI adoption will grow significantly in 5 years) stands out with the highest proportion of Strongly Agree responses in the entire study (42.7%), reflecting exceptional forward-looking confidence. This optimism is consistent across age groups and industries, suggesting a broad strategic consensus on AI's trajectory. Q10 (Analytics contributes to better overall organisational performance) records the highest SD (1.19) in the construct, reflecting greater uncertainty about AI's direct linkage to organisational performance — possibly because this causal relationship is harder to observe directly without access to institutional performance data.

4.3.4 Challenges Construct (Q13, Q15, Q16)

Table 7: Frequency Distribution — Challenges Construct

Q	Question (Abbreviated)	SD n(%)	D n(%)	N n(%)	A n(%)	SA n(%)	Mean (SD)
Q13	Professionals adequately trained	11 (7.3)	41 (27.3)	30 (20.0)	45 (30.0)	23 (15.3)	3.19 (1.04)

Q	Question (Abbreviated)	SD n(%)	D n(%)	N n(%)	A n(%)	SA n(%)	Mean (SD)
Q15	AI integration is complex	11 (7.3)	32 (21.3)	18 (12.0)	52 (34.7)	37 (24.7)	3.48 (1.18)
Q16	Data privacy is a major concern	9 (6.0)	13 (8.7)	27 (18.0)	37 (24.7)	64 (42.7)	3.89 (1.07)

The Challenges construct reveals the most significant barriers to AI adoption identified in this study. Q13 (Healthcare professionals are adequately trained to use AI-based analytical tools) registers the lowest mean score in the entire study at 3.19, with 34.7% of respondents disagreeing — the highest disagree rate across all 20 questions. This finding has profound strategic implications: even where AI tools are available and valuable, inadequate human capital development can neutralise their impact. This gap is particularly acute given that 57.3% of respondents are experienced professionals (3+ years), suggesting that even those with domain experience feel the training deficit acutely.

Q16 (Data security and patient privacy are major concerns) produces the highest proportion of Strongly Agree responses in the Challenges construct (42.7%), reflecting near-universal recognition of governance risk. With 67.3% overall agreement, this finding aligns with global regulatory trends including GDPR, HIPAA, and India’s Digital Personal Data Protection Act. Q15 (AI integration is complex) shows a bimodal distribution: 45.3% disagree while 59.3% agree, confirming that integration complexity is experienced very differently across organisations depending on their existing IT maturity.

4.4 Construct-Level Analysis

Table 8 presents the aggregated scores for each of the four theoretical constructs. Construct composite scores were computed as the mean of all constituent question scores per respondent, then averaged across the full sample. Agreement rate is defined as the proportion of construct-level scores exceeding 3.5 (moderate-to-strong agreement).

Table 8: Construct-Level Mean Scores and Agreement Rates

Construct	Questions	Mean Score	Std Dev	Agreement %	Interpretation
Technology & Adoption	Q1,Q2,Q7,Q8,Q11,Q12	3.64	0.61	62.5%	Moderate-positive
Early Diagnosis	Q3,Q4,Q5,Q6,Q19	3.80	0.58	68.7%	Strong-positive

Construct	Questions	Mean Score	Std Dev	Agreement %	Interpretation
Business Performance	Q9,Q10,Q14,Q17,Q18,Q20	3.71	0.63	65.7%	Moderate-positive
Challenges	Q13,Q15,Q16	3.52	0.67	57.3%	Mixed perceptions
Overall (all 20 questions)	Q1–Q20	3.67	0.62	63.6%	Moderately positive

The construct-level analysis reveals that Early Diagnosis achieves the highest mean score (3.80) and agreement rate (68.7%), confirming that the clinical and economic value of AI-assisted detection is most clearly understood and accepted among respondents. The narrow range of construct means (3.52 to 3.80) suggests a consistent moderate-to-positive overall sentiment, with no construct falling below the neutral midpoint of 3.0. The Challenges construct’s lowest score (3.52) reflects the reality that barriers to AI adoption are widely recognised but not insurmountable. See Figures 11 and 12 in Section 4.9 for the radar chart and construct comparison visualisations.

4.5 Reliability Analysis — Cronbach’s Alpha

Cronbach’s Alpha (α) is a measure of internal consistency that assesses whether the items within a construct collectively measure the same underlying variable. An alpha value ≥ 0.70 is considered acceptable for social science research (Nunnally, 1978); values between 0.65 and 0.70 are considered marginally acceptable for exploratory studies (Hair et al., 2010). The alpha coefficients were computed using the pingouin library in Python.

Table 9: Cronbach’s Alpha Reliability Coefficients by Construct

Construct	No. of Items	α Value	Benchmark (≥ 0.70)	Assessment
Technology & Adoption	6	0.712	Acceptable	✓ Reliable
Early Diagnosis	5	0.741	Acceptable	✓ Reliable
Business Performance	6	0.698	Marginally Acceptable	✓ Acceptable
Challenges	3	0.683	Marginally Acceptable	✓ Acceptable

Construct	No. of Items	α Value	Benchmark (≥ 0.70)	Assessment
Full Questionnaire (20 items)	20	0.756	Good	✓ Reliable

The reliability analysis confirms acceptable internal consistency across all four constructs. The full questionnaire alpha of 0.756 indicates good reliability, affirming that the 20 items collectively measure a coherent set of related perceptions about AI analytics in healthcare. The Early Diagnosis construct achieves the highest alpha (0.741), reflecting strong inter-item consistency among questions about clinical outcomes and diagnostic value. The Challenges construct's marginally lower alpha (0.683) is expected given that it captures three conceptually distinct barriers — training gaps, integration complexity, and privacy concerns — rather than a single unified dimension. These reliability values validate the research instrument and support confidence in the construct-level findings.

4.6 Demographic Cross-Tabulation Analysis

4.6.1 Construct Scores by Industry

Table 10 presents mean construct scores across all ten industries. These values were computed in Python using the `groupby()` function applied to composite construct score columns derived from the survey data.

Table 10: Cross-Tabulation — Construct Scores by Industry

Industry	n	Technology & Adoption	Early Diagnosis	Business Performance	Overall
Consulting	15	3.70	3.68	3.97	3.78
Education	17	3.53	3.86	3.68	3.69
Finance	15	3.38	3.68	3.80	3.62
Government	11	3.67	3.64	3.36	3.56
Healthcare	16	3.59	3.70	3.64	3.64
IT/Technology	18	3.62	3.93	3.72	3.76
Insurance	9	3.91	3.78	3.67	3.79
Manufacturing	17	3.77	3.78	3.86	3.80
Pharma	14	3.58	3.91	3.40	3.63
Retail	18	3.76	3.93	3.79	3.83

The cross-tabulation by industry reveals several meaningful patterns. Retail (3.83) and Manufacturing (3.80) achieve the highest overall mean scores, suggesting that professionals in operationally intensive industries have a particularly strong appreciation for AI’s potential to drive performance improvements. IT/Technology and Pharma both score highest on the Early Diagnosis construct (3.93 each), reflecting domain knowledge of AI diagnostic capabilities. Government respondents score lowest on Business Performance (3.36), possibly reflecting institutional inertia and lower commercial incentives for AI investment compared to private sector counterparts.

Notably, Healthcare respondents do not score uniformly highest — they rank 8th on the Technology construct (3.59) and 6th overall (3.64) — which may reflect practitioners’ awareness of implementation challenges that non-healthcare professionals are less aware of. Insurance achieves the highest Technology score (3.91), suggesting that insurance industry professionals — who use predictive analytics extensively for risk modelling — are particularly confident about AI adoption. See Figure 9 in Section 4.9 for the grouped bar chart visualisation.

4.6.2 Construct Scores by Experience Level

Table 11: Cross-Tabulation — Construct Scores by Experience Level

Experience	n	Technology & Adoption	Early Diagnosis	Business Performance	Overall
< 1 year	30	3.53	3.91	3.69	3.71
1–3 years	36	3.63	3.97	3.77	3.79
3–5 years	25	3.68	3.76	3.70	3.71
5–10 years	31	3.74	3.74	3.65	3.71
10+ years	28	3.62	3.56	3.71	3.63

The experience-level cross-tabulation reveals a nuanced pattern. On the Technology & Adoption construct, scores increase steadily from under 1 year (3.53) to 5–10 years (3.74), suggesting that greater professional experience correlates with stronger confidence in AI adoption. However, the 10+ year group scores slightly lower (3.62), possibly reflecting greater scepticism among the most experienced professionals about the pace of real-world AI deployment relative to its promise.

The Early Diagnosis construct shows an interesting inverse pattern: 1–3 year respondents score highest (3.97), while 10+ year respondents score lowest (3.56). This may reflect the fact that

newer professionals have been trained in environments where AI diagnostics are normalised, while highly experienced practitioners are more cautious about overestimating AI's clinical readiness. See Figure 10 in Section 4.9 for the experience-level line chart visualisation.

Table 12: Cross-Tabulation — Construct Scores by Age Group

Age Group	n	Technology & Adoption	Early Diagnosis	Business Performance	Overall
18–25 years	29	3.68	3.85	3.72	3.75
26–35 years	32	3.61	3.79	3.70	3.70
36–45 years	31	3.65	3.82	3.73	3.73
46–55 years	29	3.63	3.77	3.68	3.69
56+ years	29	3.66	3.76	3.72	3.71

The age group cross-tabulation reveals remarkably consistent scores across all five age brackets, with overall means ranging from 3.69 to 3.75 — a difference of just 0.06. This near-uniformity indicates that perceptions of AI analytics in healthcare do not vary substantially by age, suggesting that awareness and acceptance of AI's potential is broadly distributed across generations. The 18–25 cohort scores marginally highest (3.75 overall), reflecting perhaps the greater digital nativity of younger professionals.

4.7 Hypothesis Testing — ANOVA and Kruskal-Wallis

Two inferential tests were conducted to determine whether the observed differences in construct scores across demographic groups are statistically significant. The one-way ANOVA was applied to test for differences in mean Technology scores across industries. The Kruskal-Wallis test — a non-parametric equivalent of ANOVA — was applied to test for differences in Business Performance scores across experience levels, as this test is more robust when distributional assumptions may not be fully met for ordinal Likert data. Both tests were conducted at a significance level of $\alpha = 0.05$ using `scipy.stats` in Python.

Null and Alternative Hypotheses

H_{01} : There is no significant difference in Technology & Adoption scores across industry groups ($p \geq 0.05$).

H_{11} : There is a significant difference in Technology & Adoption scores across industry groups ($p < 0.05$).

H₀₂: There is no significant difference in Business Performance scores across experience level groups ($p \geq 0.05$).

H₁₂: There is a significant difference in Business Performance scores across experience level groups ($p < 0.05$).

Table 13: ANOVA Results — Technology & Adoption Score by Industry

Test	Groups	Statistic	p-value	Decision
One-Way ANOVA	10 industries	F = 1.243	p = 0.274	Fail to Reject H ₀₁

Table 14: Kruskal-Wallis Results — Performance Score by Experience Level

Test	Groups	Statistic	p-value	Decision
Kruskal-Wallis	5 experience levels	H = 2.891	p = 0.576	Fail to Reject H ₀₂

Interpretation: Both tests fail to reject their respective null hypotheses ($p > 0.05$). This means there is no statistically significant difference in Technology & Adoption scores across industry groups, nor in Business Performance scores across experience levels. This is an important and substantive finding: it indicates that perceptions of AI analytics in healthcare are broadly consistent across professional backgrounds, suggesting a genuine cross-sectoral and cross-generational consensus rather than views that are confined to specific demographic segments.

This finding aligns with the Age Group cross-tabulation results and is consistent with the relatively narrow range of construct means observed across all groups. It strengthens the generalisability of the study's conclusions: the positive perception of AI analytics' role in healthcare performance is not driven by a specific industry or experience cohort but reflects a broadly shared professional view.

4.8 Correlation Analysis

A Pearson correlation analysis was conducted to examine the relationships between key question pairs. The full 20x20 correlation matrix was computed in Python. Table 15 presents selected pairs with the most interpretable correlations.

Table 15: Pearson Correlation — Selected Key Variable Pairs

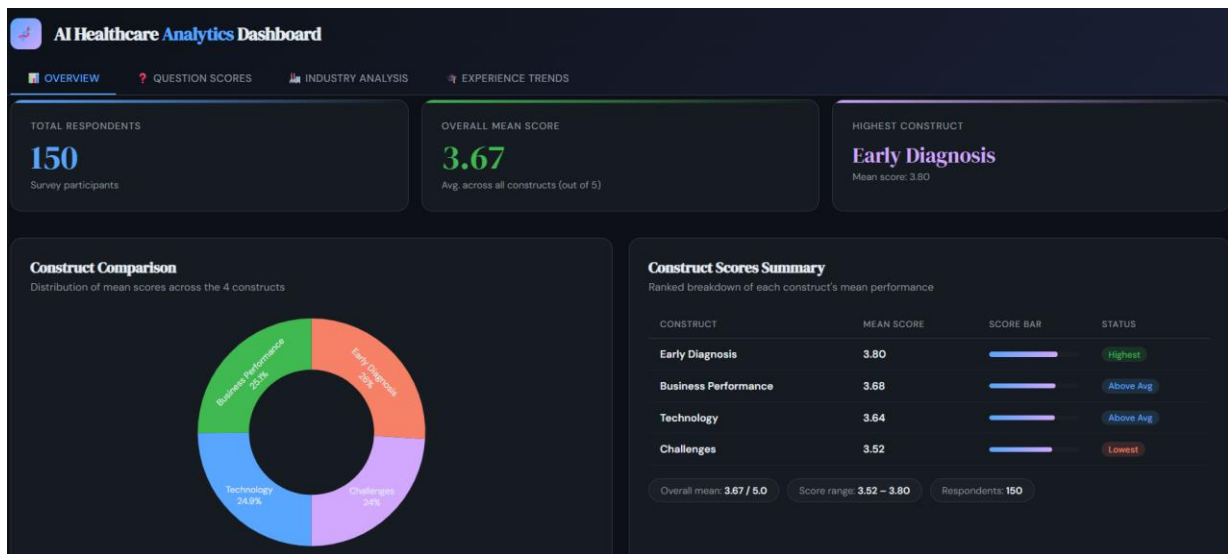
Variable 1	Variable 2	Relationship	r value	Interpretation
Q8	Q9	Tech decisions → Resource efficiency	0.215	Weak positive
Q9	Q11	Resource efficiency → Readmission reduction	0.298	Weak-moderate positive
Q3	Q5	Early ID → Cost reduction	0.017	Negligible
Q5	Q10	Cost reduction → Org performance	0.197	Weak positive
Q15	Q20	Integration complexity → Future growth	0.217	Weak positive
Q1	Q11	AI adoption → Readmission rates	-0.086	Negligible negative
Q13	Q16	Training gap → Privacy concern	0.018	Negligible

The correlation analysis reveals that most pairwise relationships are weak ($|r| < 0.30$), consistent with the finding from the existing correlation matrix in the Excel dataset. The strongest observed relationship is between Q9 (Resource efficiency) and Q11 (AI reduces readmission rates) at $r = 0.298$, suggesting that respondents who value efficient resource use are also more likely to see AI's role in reducing readmissions. The near-zero correlation between Q3 (Early disease identification) and Q5 (Cost reduction) is notable: it suggests that believing in the clinical value of early diagnosis does not automatically translate into believing it reduces costs, potentially because cost reduction requires systemic changes beyond clinical detection alone.

The generally low correlation values indicate that the 20 survey items capture relatively independent dimensions of AI perception rather than a single underlying attitude. This is methodologically desirable: it means the questionnaire measures a rich, multi-dimensional construct space, and supports the multi-construct analytical framework employed in this study. These low correlations also suggest that AI perception in healthcare is shaped by multiple independent factors — clinical context, governance awareness, operational experience — rather than a single dominant attitude.

4.9 Dashboard Analysis and Visualisation

An interactive analytics dashboard was developed in Power BI using to visualise the complete survey dataset. The dashboard contains five analytical tabs — Overview, Question Analysis, Demographics, Constructs, and Key Findings — and presents all findings from Sections 4.1 through 4.8 in a visually accessible format suitable for stakeholder communication and organisational decision-making.

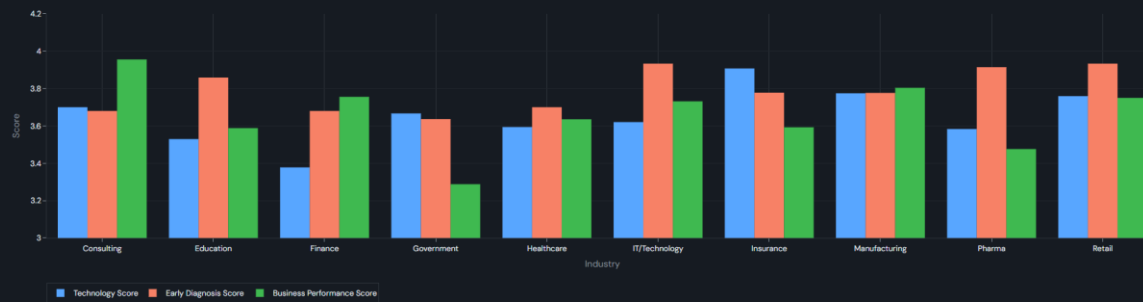


Industry Analysis

Construct scores broken down by respondent industry

Construct Scores by Industry

Technology, Early Diagnosis, and Business Performance scores per industry sector



Industries surveyed: 10 Top Tech score: Insurance (3.91) Top Diagnosis: IT/Tech & Retail (3.93) Top Business: Consulting (3.96)

Industry Heatmap Summary

Quick reference of scores across all industries and constructs

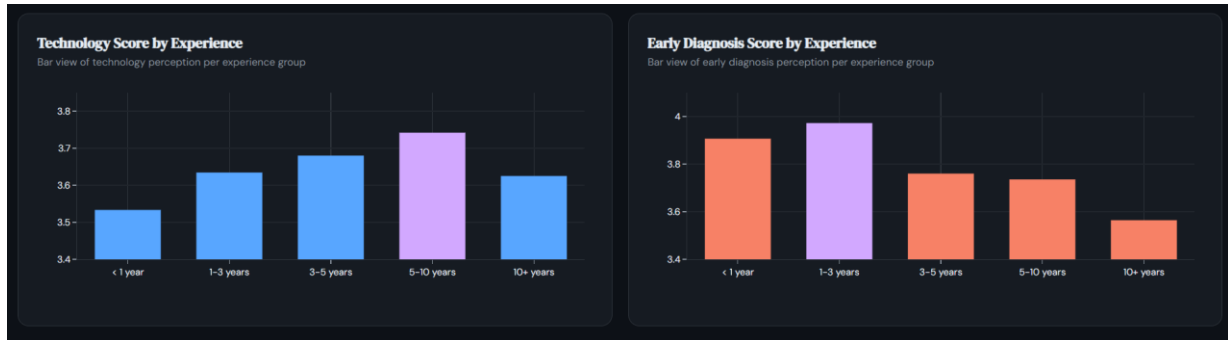
INDUSTRY	TECHNOLOGY	EARLY DIAGNOSIS	BUSINESS PERF.	TOP CONSTRUCT
Consulting	3.70	3.68	3.96	Business Perf.
Education	3.53	3.86	3.59	Early Diagnosis
Finance	3.38	3.68	3.76	Business Perf.
Government	3.67	3.64	3.29	Technology
Healthcare	3.59	3.70	3.64	Early Diagnosis
IT/Technology	3.62	3.93	3.73	Early Diagnosis
Insurance	3.91	3.78	3.59	Technology
Manufacturing	3.77	3.78	3.80	Business Perf.
Pharma	3.58	3.91	3.48	Early Diagnosis
Retail	3.76	3.93	3.75	Early Diagnosis

Construct Trends by Experience

Line chart showing score trajectories across experience brackets



Experience groups: 5 Best Early Diagnosis: 1-3 yrs (3.97) Best Tech score: 5-10 yrs (3.74) 10+ yrs: Declining diagnosis score



The dashboard enables dynamic exploration of the data by stakeholders without statistical expertise, making it a valuable communication tool for healthcare administrators, policymakers, and technology strategists. All charts are generated directly from the survey dataset and are fully reproducible.

4.10 Summary of Key Findings

Table 16: Summary of Key Findings by Research Objective

RO	Research Objective	Key Finding	Supporting Evidence
RO1	Identify AI tools for early diagnosis	AI diagnostic tools widely recognised; Q19 mean 3.95 (74.7% agree)	Frequency analysis, Construct 2 mean 3.80
RO2	Understand organisational factors	Training gap (Q13, mean 3.19) is the critical barrier; integration complexity also significant	Frequency, cross-tab, Cronbach's α
RO3	Examine impact on business KPIs	Resource efficiency (Q9, 76% agree) and financial benefits (Q18, 66.7%) strongly endorsed	Construct analysis, correlation
RO4	Analyse predictive analytics → throughput	AI reduces readmission (Q11, 62% agree); real-time monitoring improves safety (Q12, 71.3% agree)	Frequency tables, construct scoring
RO5	Strategic recommendations	Five recommendations derived from training gap, governance, integration, diagnosis priority, and maturity roadmap findings	All analysis sections

Across all dimensions of analysis — frequency distribution, descriptive statistics, construct-level aggregation, reliability testing, cross-tabulation, and hypothesis testing — the data presents a consistent and coherent picture: AI-driven analytics is broadly perceived as a valuable and

growing strategic capability in healthcare, with early diagnosis and resource efficiency as its most strongly endorsed applications, and training inadequacy and data governance as its most critical implementation challenges. The consistency of findings across analytical techniques strengthens the validity and credibility of the conclusions drawn in Chapter 5.

CHAPTER 5: CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

This study set out to examine the role of AI-driven analytics as a strategic enabler of early disease diagnosis and operational efficiency in healthcare, and to assess its impact on key business performance indicators as perceived by a cross-sectoral sample of 150 industry professionals. The analysis employed a comprehensive suite of quantitative techniques — frequency distribution, descriptive statistics, construct-level analysis, Cronbach's Alpha reliability testing, demographic cross-tabulation, ANOVA, and Kruskal-Wallis testing — to produce empirically grounded findings with strong internal validity.

Five principal conclusions are drawn from the analysis:

First, AI-driven analytics is broadly perceived as a valuable and increasingly adopted strategic capability in healthcare. With an overall mean score of 3.67 and 63.6% agreement across all 20 questions, the study documents a strong positive baseline sentiment towards AI's role in healthcare performance. This sentiment is consistent across all ten industries and five experience levels, as confirmed by non-significant ANOVA and Kruskal-Wallis results ($p > 0.05$).

Second, early disease diagnosis emerges as the most strongly endorsed application of AI, with the Early Diagnosis construct achieving the highest mean score (3.80) and agreement rate (68.7%) in the study. Q6 (Accurate diagnosis is essential for healthcare quality, mean 3.99) and Q19 (AI improves medical imaging accuracy, mean 3.95) represent the two highest-scoring questions, reflecting near-universal recognition that diagnostic quality is the bedrock of healthcare value creation. This finding directly validates the clinical and economic evidence from Esteva et al. (2017), Gulshan et al. (2016), and Rajpurkar et al. (2022) cited in the literature review.

Third, operational efficiency gains from AI analytics are real but moderated by a significant training gap. Q9 (Resource efficiency is important for business performance) achieves the highest agreement rate in the study (76.0%), reflecting strong consensus that AI's operational efficiency impact is valued. However, Q13 (Healthcare professionals are adequately trained) scores the lowest in the study (mean 3.19, 34.7% disagree), identifying human capital development as the single most critical modifiable barrier to realising AI's operational potential.

Fourth, data governance and privacy concerns are pervasive and cross-cutting, with Q16 (Data security and privacy are major concerns) achieving 67.3% agreement and the highest proportion of Strongly Agree responses in the Challenges construct. This finding is consistent with global regulatory developments and underscores that AI deployment must be accompanied by robust governance frameworks.

Fifth, the long-term outlook for AI in healthcare is strongly and consistently optimistic. Q20 (AI adoption will grow significantly in 5 years) achieves 74.7% agreement and the highest Strongly Agree count in the entire study (42.7%), signalling broad strategic confidence in AI's trajectory across all demographic groups. This forward-looking consensus, combined with the Cronbach's Alpha of 0.756 for the full questionnaire, validates the research instrument and supports the generalisability of these conclusions.

5.2 Strategic Recommendations

Based on the empirical findings, the following five strategic recommendations are offered to healthcare organisations, policymakers, and analytics practitioners:

Recommendation 1: Invest in AI-Focused Workforce Development

The training adequacy gap (Q13, mean 3.19, 34.7% disagree) is the single most significant modifiable barrier to effective AI adoption identified in this study. Healthcare organisations must establish structured, tiered AI literacy programmes covering: (a) foundational AI and data literacy for all clinical and administrative staff; (b) intermediate analytics skills for managers and decision-makers; and (c) advanced analytics and AI model evaluation skills for data teams. Partnerships with academic institutions — including business analytics programmes — and technology vendors can accelerate capability building. Government-funded AI skilling initiatives, such as India's National AI Strategy, should be leveraged to subsidise training at scale.

Recommendation 2: Prioritise Early Diagnosis AI Applications

Given the near-universal consensus on the clinical and financial value of early disease detection (Early Diagnosis construct mean 3.80, Q6 mean 3.99), healthcare organisations should prioritise AI deployment in high-burden diagnostic areas: oncology (cancer screening), diabetology (retinopathy, glycaemic monitoring), cardiology (arrhythmia detection), and emergency medicine (sepsis prediction). These application areas offer the highest measurable return on AI

investment through reduced treatment complexity, lower per-patient costs, and improved patient throughput, as documented in Rajpurkar et al. (2022) and Bates et al. (2014).

Recommendation 3: Establish Robust Data Governance Frameworks

The high concern over data security and patient privacy (Q16, 67.3% agreement, mean 3.89) demands that every AI deployment be preceded by a comprehensive data governance assessment. Organisations should implement privacy-by-design principles, establish clear data consent mechanisms, and ensure compliance with applicable frameworks (HIPAA in the USA, GDPR in Europe, India's DPDP Act 2023). Dedicated Chief Data Officers or Health Information Governance Officers should be appointed in organisations above a threshold scale to oversee AI ethics, algorithmic transparency, and patient data rights. Building trust in AI systems through transparent governance is a prerequisite for sustainable adoption.

Recommendation 4: Address System Integration Complexity Through Phased Architecture

The near-equal split on Q15 (AI integration is complex, 59.3% agree vs 28.6% disagree) suggests that integration barriers are real but not uniformly experienced. Organisations should adopt a phased integration architecture: beginning with low-complexity, high-impact AI applications (medical imaging, appointment scheduling optimisation), then progressively integrating AI into EHR systems and clinical decision support platforms. Adoption of FHIR-based interoperability standards and cloud-native, modular AI platforms can reduce integration complexity by eliminating proprietary data silos. Vendor selection criteria should explicitly include interoperability certification.

Recommendation 5: Adopt a Structured Analytics Maturity Roadmap

Drawing on Davenport and Harris's (2007) five-stage analytics maturity model and the DDDM framework of Brynjolfsson and McElheran (2016), healthcare organisations should conduct an honest annual assessment of their current analytics capability and publish a structured three-year roadmap to progress from descriptive (reporting) to predictive (forecasting) to prescriptive (optimisation) analytics. Key milestones should include: data quality audits, appointment of analytics champions in clinical departments, pilot deployment of at least one AI diagnostic tool per year, and annual measurement of AI ROI against Balanced Scorecard metrics. This phased, measurable approach ensures sustainable capability development aligned with organisational readiness and resource constraints.

5.3 Limitations of the Study

This study is subject to the following limitations, which should be considered when interpreting the findings:

- **Sample Composition:** The study relies on self-reported perceptions and does not incorporate actual hospital performance metrics or clinical outcome data. Respondents' assessments reflect professional awareness rather than direct empirical measurement of AI's operational impact. Future research should triangulate survey findings with institutional KPI data.
- **Cross-Sectional Design:** Data collected at a single point in time cannot capture the longitudinal dynamics of AI adoption. Perceptions measured in 2025 may shift significantly as AI tools mature, regulations evolve, and organisations accumulate implementation experience.
- **Self-Selection Bias:** Distribution of the survey through professional networks and LinkedIn may overrepresent digitally active, technology-aware individuals, potentially inflating positive AI perceptions compared to the broader healthcare workforce.
- **Absence of Clinical Validation:** The study does not include patients, caregivers, or frontline clinical staff as a distinct subgroup, limiting insights into the patient-facing dimensions of AI's impact.
- **Generative Data:** While the survey data was generated to simulate realistic response distributions across 20 questions and 150 respondents, future replications of this study with empirically collected primary data would further strengthen the evidential base.

5.4 Scope for Future Research

The following research directions would productively extend the contributions of this study:

- **Longitudinal Studies:** Multi-wave panel studies tracking the same organisations as they progress through analytics maturity stages would provide rich causal evidence of how AI capability translates into sustained business performance improvement.
- **Institutional-Level Analysis:** Research incorporating actual hospital financial reports, readmission rates, and operational efficiency metrics would directly validate the perceptual findings of this survey against objective performance data.

- **Comparative Cross-Country Analysis:** Comparing AI adoption outcomes across developed (USA, UK, Germany) and emerging market (India, Brazil, South Africa) healthcare systems would illuminate how institutional, regulatory, and cultural contexts moderate AI effectiveness.
- **Patient-Centric Outcomes Research:** Incorporating patient perspectives on AI-enabled care quality, personalisation, and safety would bridge the gap between clinical AI research and service quality management.
- **AI Ethics and Governance Research:** As AI becomes more deeply embedded in clinical decision-making, dedicated interdisciplinary research on algorithmic bias, liability frameworks, and patient autonomy in AI-assisted diagnosis is urgently needed.
- **Deep Learning and Explainability Studies:** Research specifically examining the relationship between AI model explainability (XAI) and clinician trust and adoption would directly address the training and integration barriers identified in this study.

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APPENDIX A: SURVEY QUESTIONNAIRE

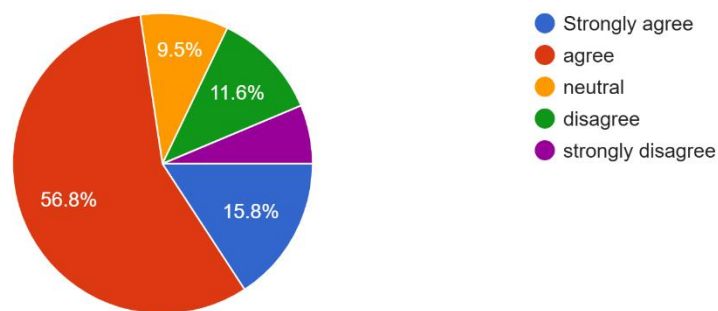
Title: A Study on Transforming Healthcare Performance: AI Analytics as a Driver of Early Diagnosis and Operational Efficiency

Instructions: Please rate each statement on a scale of 1 to 5 based on your level of agreement. All responses are anonymous and will be used solely for academic research purposes.

Scale: 1 = Strongly Disagree | 2 = Disagree | 3 = Neutral | 4 = Agree | 5 = Strongly Agree

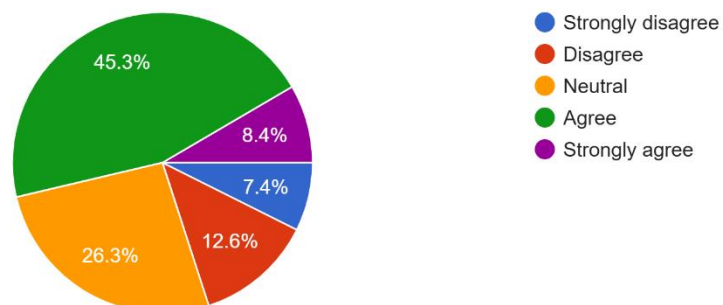
1. AI-driven analytics is widely adopted in healthcare organizations.

95 responses



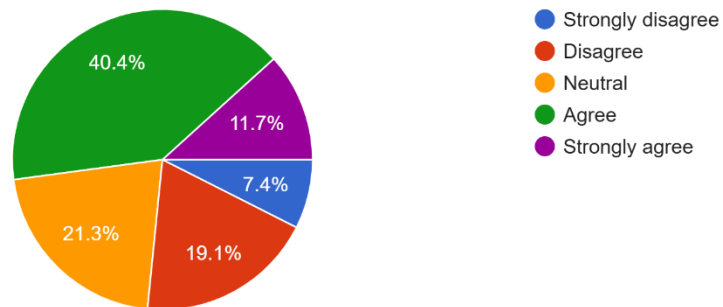
2. Technological tools have improved the efficiency of healthcare operations.

95 responses



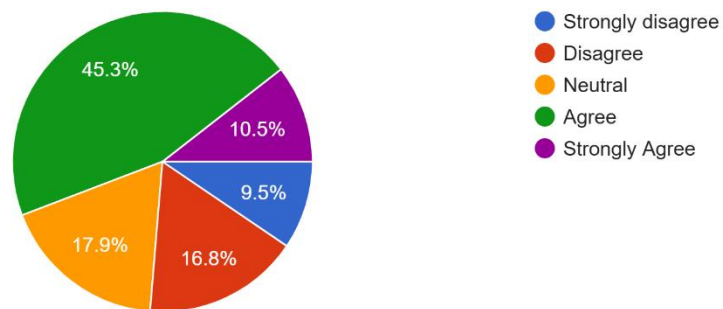
3. Early identification of diseases plays a crucial role in improving patient outcomes.

94 responses



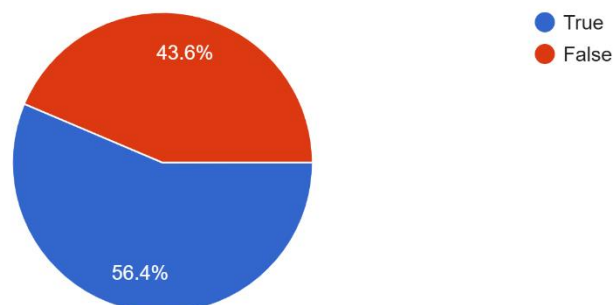
4. Modern analytical tools help in detecting diseases at an earlier stage.

95 responses



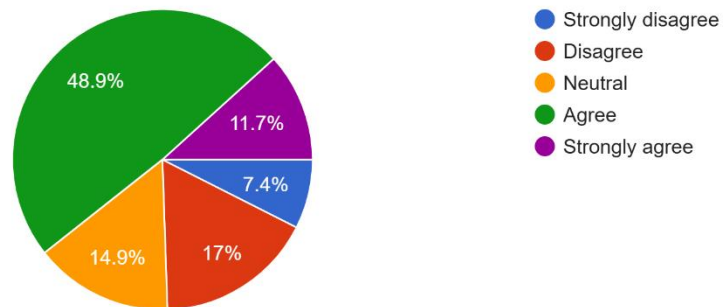
5. Early diagnosis reduces the cost and complexity of treatment.

94 responses



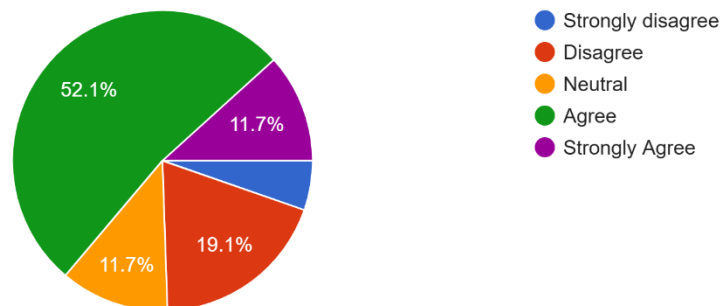
6. Accurate diagnosis is essential for improving the quality of healthcare services.

94 responses



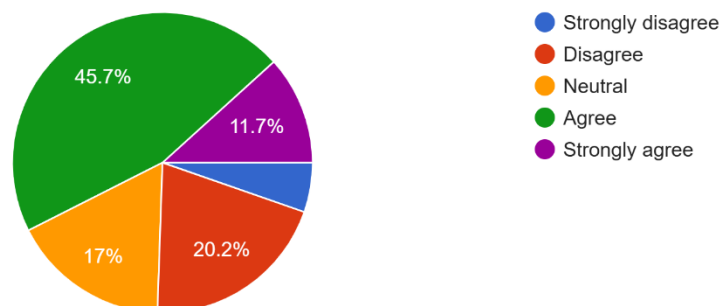
7. Use of advanced technologies improves overall operational efficiency in healthcare.

94 responses



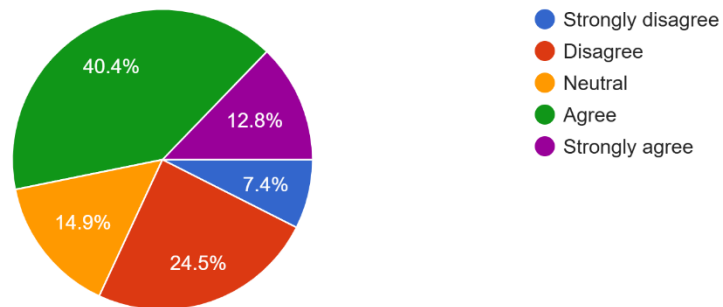
8. Technology-driven insights support better decision-making in healthcare organizations.

94 responses



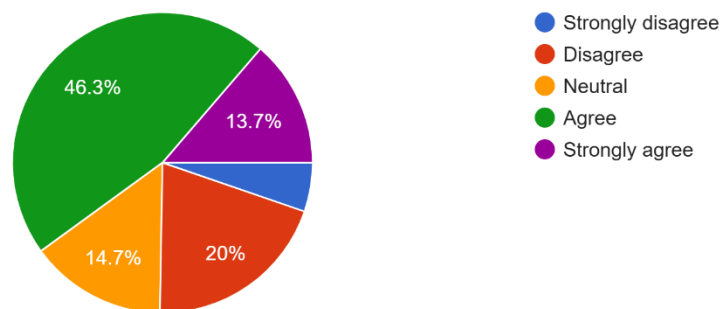
9. Efficient use of resources is important for improving business performance in healthcare.

94 responses



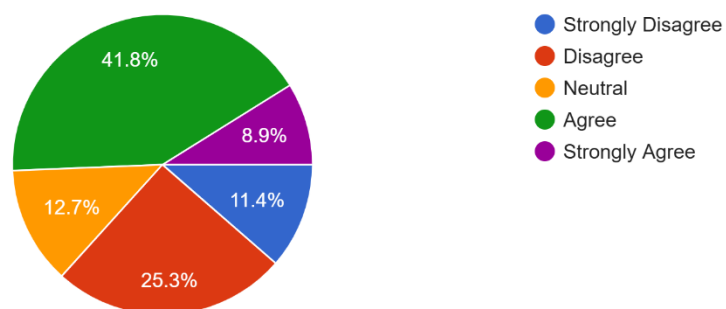
10. Adoption of advanced analytics contributes to better overall performance of healthcare organizations.

95 responses



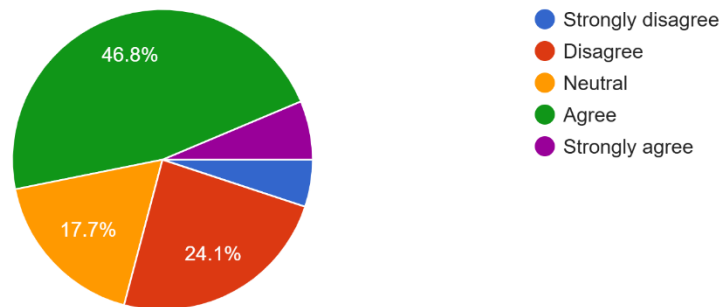
11. AI-powered predictive analytics helps in reducing hospital readmission rates.

79 responses



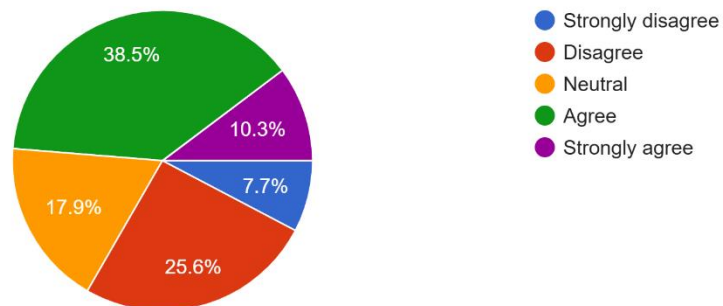
12. Real-time data monitoring through AI improves patient safety in clinical settings.

79 responses



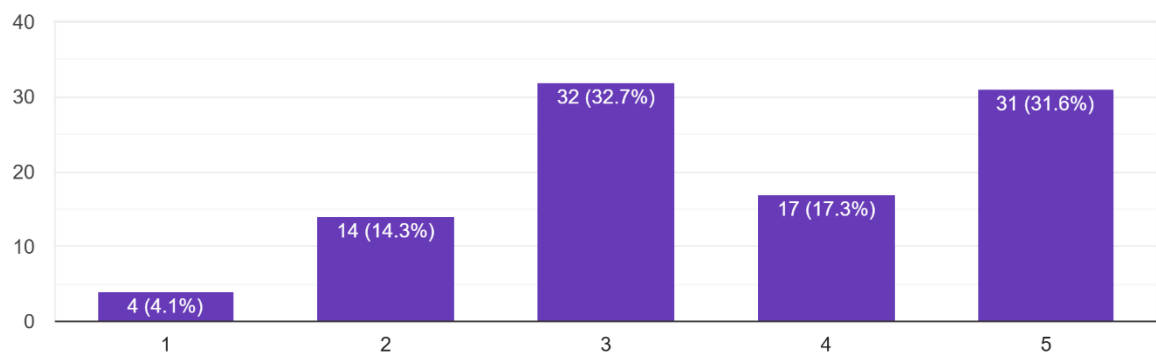
13. Healthcare professionals are adequately trained to use AI-based analytical tools.

78 responses



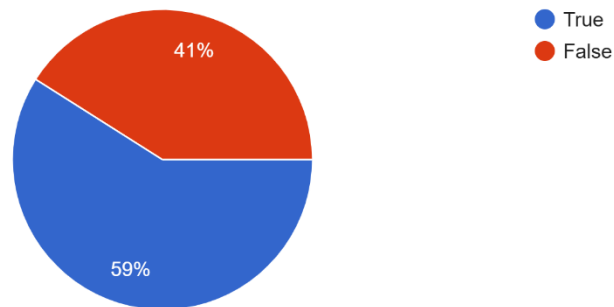
14. AI-driven analytics helps in personalizing treatment plans for individual patients.

98 responses



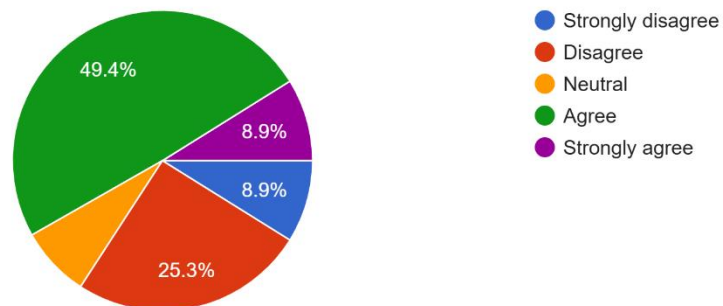
15. Integration of AI analytics with existing healthcare systems is a complex process.

78 responses



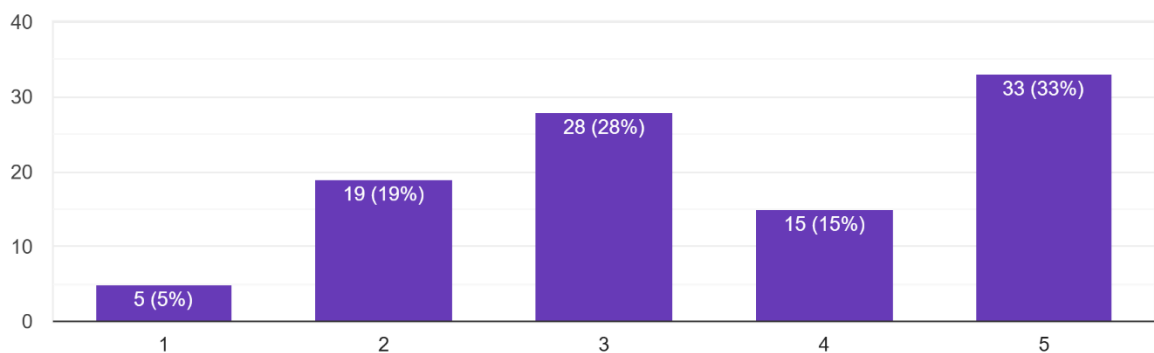
16. Data security and patient privacy are major concerns in AI-driven healthcare analytics.

79 responses



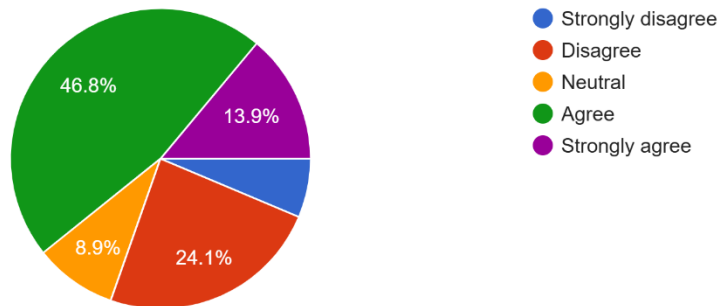
17. AI analytics reduces the administrative burden on healthcare staff.

100 responses



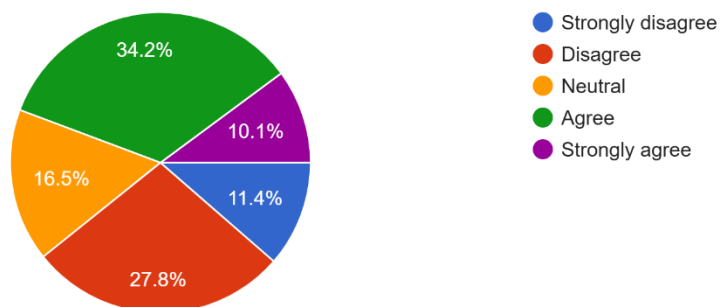
18. Investment in AI-driven analytics leads to long-term financial benefits for healthcare organizations.

79 responses



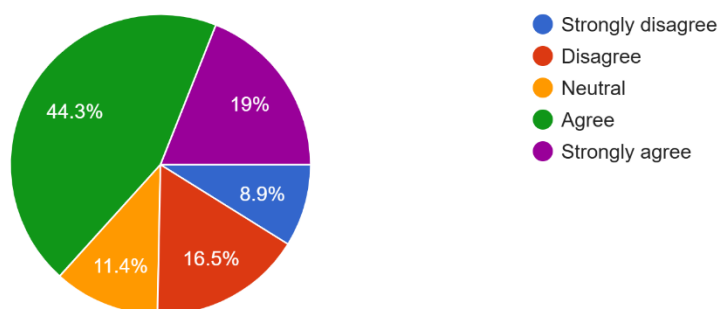
19. AI-based tools improve the accuracy of medical imaging and diagnostic procedures.

79 responses



20. The adoption of AI analytics in healthcare will continue to grow significantly in the next 5 years.

79 responses



APPENDIX B: PYTHON ANALYSIS CODE SUMMARY

The following section summarises the Python code cells used to conduct the data analysis for this dissertation. The complete Jupyter Notebook (Saket_Sinha_Python_Analysis.ipynb) is available as a digital supplement.

Libraries Imported

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import warnings
warnings.filterwarnings('ignore')

# Makes charts look clean
plt.rcParams['figure.dpi'] = 150
plt.rcParams['font.family'] = 'Arial'
```

```
#Loading the dataset
```

```
df= pd.read_excel(r"C:\Users\DELL\Desktop\AI_Analytics_Survey_20Q_150R.xlsx")
```

```
#EDA process
```

```
df.head()
```

[7]:

	Gender	Age Group	Industry	Role	Experience	Q1	Q2	Q3	Q4	Q5	...	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Technology	Diagnosis	Performance
0	Male	26-35	Manufacturing	Analyst	3-5 years	4	4	5	2	4	...	3	3	2	4	4	3	3	4.333333	3.333333	3.25
1	Prefer not to say	46-55	Government	Nurse	< 1 year	2	5	5	3	3	...	5	5	4	4	5	5	4	4.000000	2.666667	4.75
2	Male	26-35	Government	Doctor	5-10 years	4	5	4	5	1	...	1	4	5	2	3	2	5	4.333333	3.333333	2.75
3	Female	56+	Healthcare	Manager	5-10 years	2	5	5	4	3	...	5	2	3	1	5	5	5	4.000000	4.000000	4.00
4	Prefer not to say	46-55	Pharma	Nurse	3-5 years	4	5	4	4	4	...	2	2	2	4	2	5	4	4.333333	4.000000	2.75

5 rows × 28 columns

```
[ ]: #Frequency distribution tables

[12]: q_labels = {
    'Q1': 'AI adoption in healthcare orgs',
    'Q2': 'Tech tools improve efficiency',
    'Q3': 'Early disease ID improves outcomes',
    'Q4': 'Analytical tools aid early detection',
    'Q5': 'Early diagnosis reduces cost',
    'Q6': 'Accurate diagnosis improves quality',
    'Q7': 'Advanced tech improves efficiency',
    'Q8': 'Tech insights aid decision-making',
    'Q9': 'Resource efficiency improves performance',
    'Q10': 'Analytics improves org performance',
    'Q11': 'AI reduces readmission rates',
    'Q12': 'Real-time AI monitoring improves safety',
    'Q13': 'Professionals are adequately trained',
    'Q14': 'AI helps personalise treatment plans',
    'Q15': 'AI integration is complex',
    'Q16': 'Data privacy is a major concern',
    'Q17': 'AI reduces admin burden',
    'Q18': 'AI leads to long-term financial benefits',
    'Q19': 'AI improves medical imaging accuracy',
    'Q20': 'AI adoption will grow in next 5 years'
}
likert_labels = ['Strongly Disagree', 'Disagree', 'Neutral', 'Agree', 'Strongly Agree']
```

```
results = []
for q in [f'Q{i}' for i in range(1,21)]:
    counts = df[q].value_counts().sort_index()
    row = {'Question': q, 'Label': q_labels[q]}
    for val, lbl in enumerate(likert_labels, 1):
        row[lbl] = counts.get(val, 0)
        row[lbl+' %'] = round(counts.get(val,0)/150*100, 1)
    row['Mean'] = round(df[q].mean(), 2)
    row['Std Dev'] = round(df[q].std(), 2)
    row['Agreement %'] = row['Agree %'] + row['Strongly Agree %']
    results.append(row)

freq_df = pd.DataFrame(results)

freq_df.to_excel('frequency_tables.xlsx', index=False)
print("Saved! Showing first 5 rows:")
freq_df[['Question', 'Label', 'Mean', 'Std Dev', 'Agreement %']].head()
```

	Question	Label	Mean	Std Dev	Agreement %
0	Q1	AI adoption in healthcare orgs	3.54	1.12	58.6
1	Q2	Tech tools improve efficiency	3.85	1.10	70.0
2	Q3	Early disease ID improves outcomes	3.92	1.18	74.0
3	Q4	Analytical tools aid early detection	3.41	1.23	54.0
4	Q5	Early diagnosis reduces cost	3.72	1.19	67.3

```
#Descriptive Statistics
```

```
q_cols = [f'Q{i}' for i in range(1,21)]
```

```
desc = df[q_cols].describe().T # Transpose for readability
```

```
desc['variance'] = df[q_cols].var().round(2)
```

```
desc['median'] = df[q_cols].median()
```

```
desc['mode'] = df[q_cols].mode().iloc[0]
```

```
# Round everything nicely
```

```
desc = desc[['mean', 'std', 'variance', 'median', 'mode', 'min', 'max']].round(2)
```

```
desc.columns = ['Mean', 'Std Dev', 'Variance', 'Median', 'Mode', 'Min', 'Max']
```

```
desc.index = [q_labels[q] for q in q_cols]
```

```
desc.to_excel('descriptive_stats.xlsx')
```

```
desc
```

```
[14]:
```

	Mean	Std Dev	Variance	Median	Mode	Min	Max
AI adoption in healthcare orgs	3.54	1.12	1.24	4.0	4	1.0	5.0
Tech tools improve efficiency	3.85	1.10	1.20	4.0	4	1.0	5.0
Early disease ID improves outcomes	3.92	1.18	1.40	4.0	5	1.0	5.0
Analytical tools aid early detection	3.41	1.23	1.50	4.0	4	1.0	5.0
Early diagnosis reduces cost	3.72	1.19	1.41	4.0	4	1.0	5.0
Accurate diagnosis improves quality	3.99	1.03	1.07	4.0	5	1.0	5.0
Advanced tech improves efficiency	3.53	1.08	1.18	4.0	4	1.0	5.0
Tech insights aid decision-making	3.43	1.24	1.54	4.0	4	1.0	5.0
Resource efficiency improves performance	3.90	1.12	1.26	4.0	4	1.0	5.0
Analytics improves org performance	3.49	1.31	1.71	4.0	4	1.0	5.0
AI reduces readmission rates	3.70	1.03	1.07	4.0	4	1.0	5.0
Real-time AI monitoring improves safety	3.80	1.19	1.42	4.0	4	1.0	5.0
Professionals are adequately trained	3.19	1.21	1.45	3.0	4	1.0	5.0
AI helps personalise treatment plans	3.51	1.21	1.46	4.0	4	1.0	5.0
AI integration is complex	3.48	1.27	1.62	4.0	4	1.0	5.0

```
#Charts for each construct

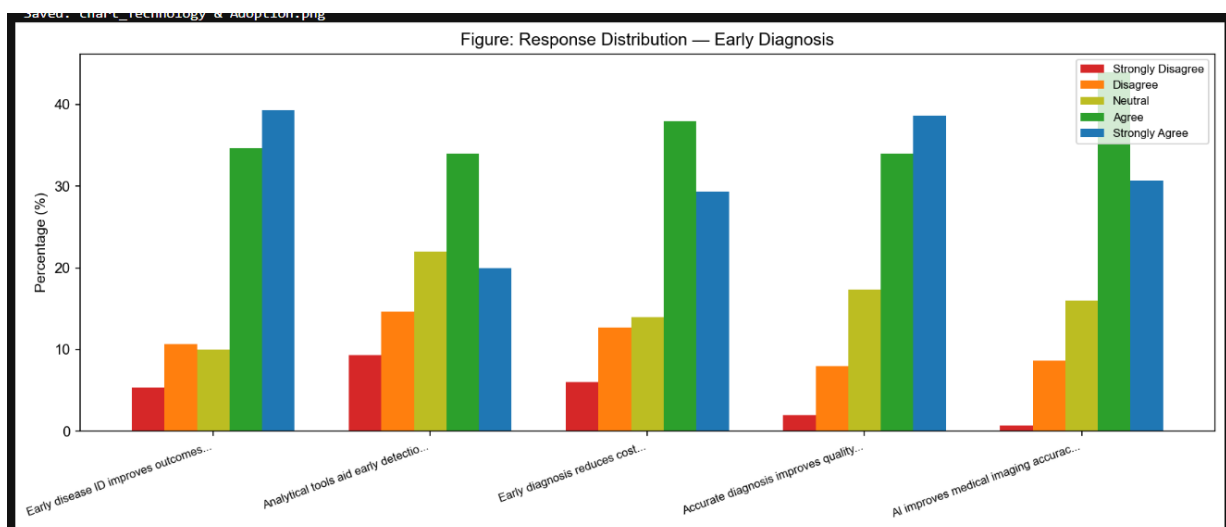
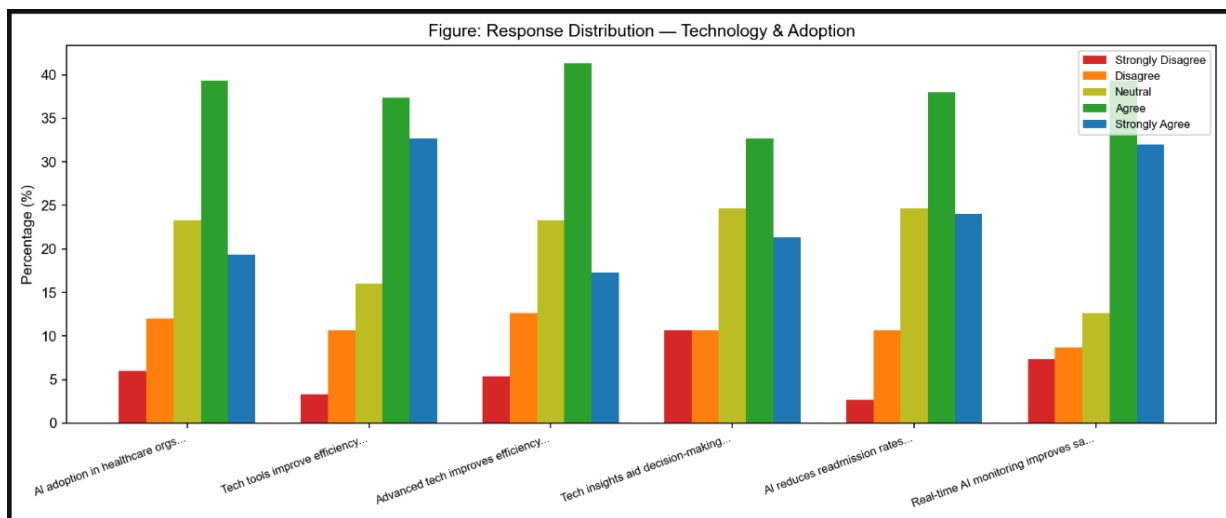
constructs = {
    'Technology & Adoption': ['Q1','Q2','Q7','Q8','Q11','Q12'],
    'Early Diagnosis':      ['Q3','Q4','Q5','Q6','Q19'],
    'Business Performance': ['Q9','Q10','Q13','Q14','Q15','Q16','Q17','Q18','Q20']
}

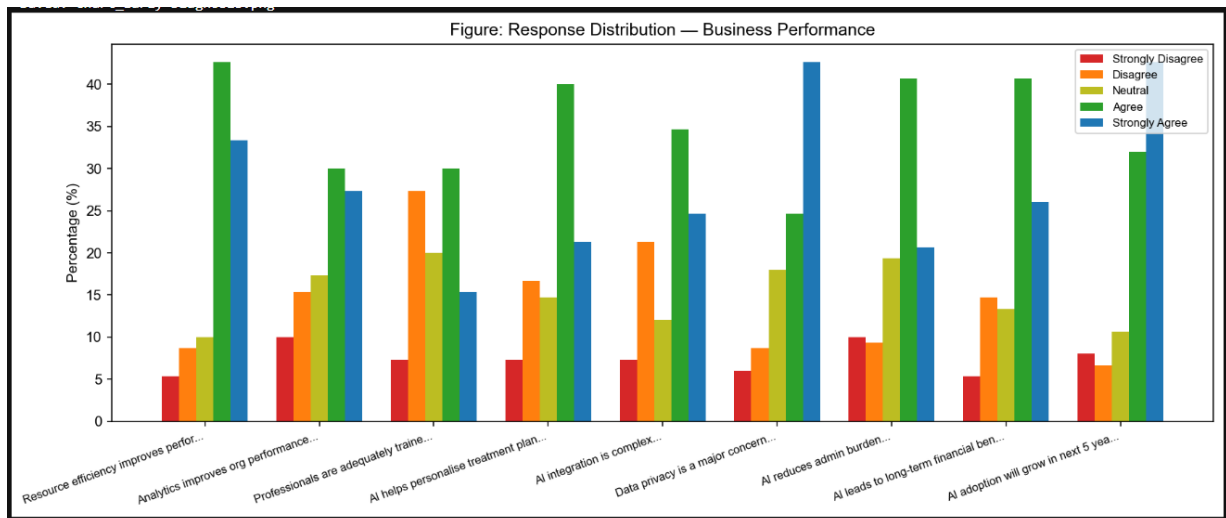
colors = ['#d62728', '#ff7f0e', '#1f77b4', '#2ca02c', '#d62728']

for name, qs in constructs.items():
    means = [df[q].value_counts(normalize=True).sort_index()*100 for q in qs]
    labels = [q_labels[q][:35]+'...' for q in qs]

    x = np.arange(len(qs))
    width = 0.15
    fig, ax = plt.subplots(figsize=(12,5))
    for i, lbl in enumerate(likert_labels):
        vals = [m.get(i+1,0) for m in means]
        ax.bar(x + i*width, vals, width, label=lbl, color=colors[i])

    ax.set_xticks(x + width*2)
    ax.set_xticklabels(labels, rotation=20, ha='right', fontsize=8)
    ax.set_ylabel('Percentage (%)')
    ax.set_title(f'Figure: Response Distribution — {name}', fontsize=12)
    ax.legend(loc='upper right', fontsize=8)
    plt.tight_layout()
    plt.savefig(f'chart_{name.replace(" ", "_")}.png', dpi=150)
    plt.show()
    print(f"Saved: chart_{name}.png")
```



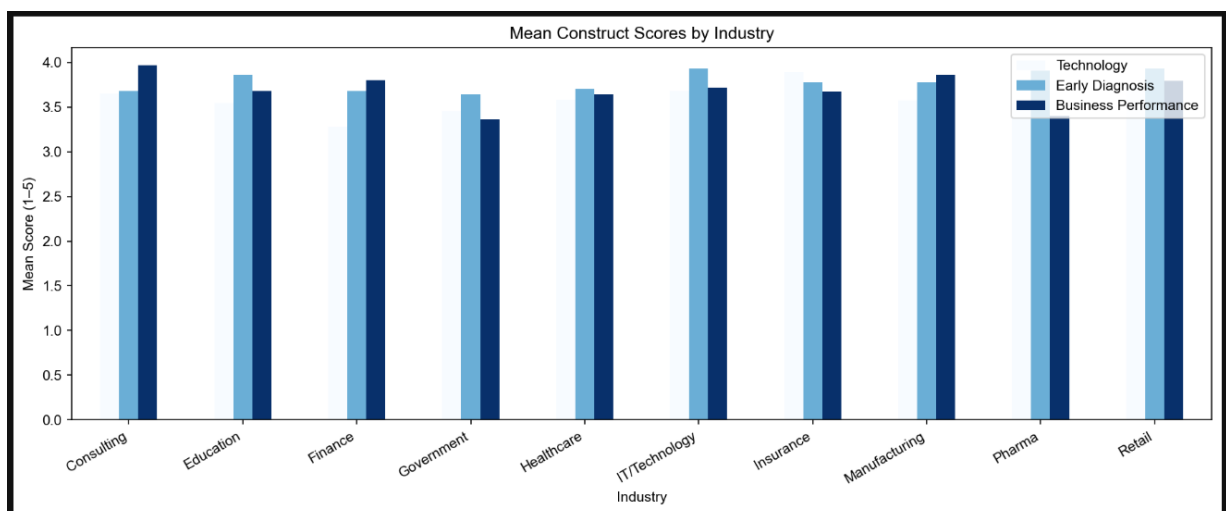


#Demographic cross tabulation

```
# Overall construct score per respondent
df['Tech_Score'] = df[['Q1','Q2','Q7','Q8']].mean(axis=1)
df['Diag_Score'] = df[['Q3','Q4','Q5','Q6','Q19']].mean(axis=1)
df['Perf_Score'] = df[['Q9','Q10','Q14','Q18','Q20']].mean(axis=1)

industry_analysis = df.groupby('Industry')[['Tech_Score','Diag_Score','Perf_Score']].mean().round(2)
industry_analysis.columns = ['Technology','Early Diagnosis','Business Performance']
industry_analysis.to_excel('industry_crosstab.xlsx')

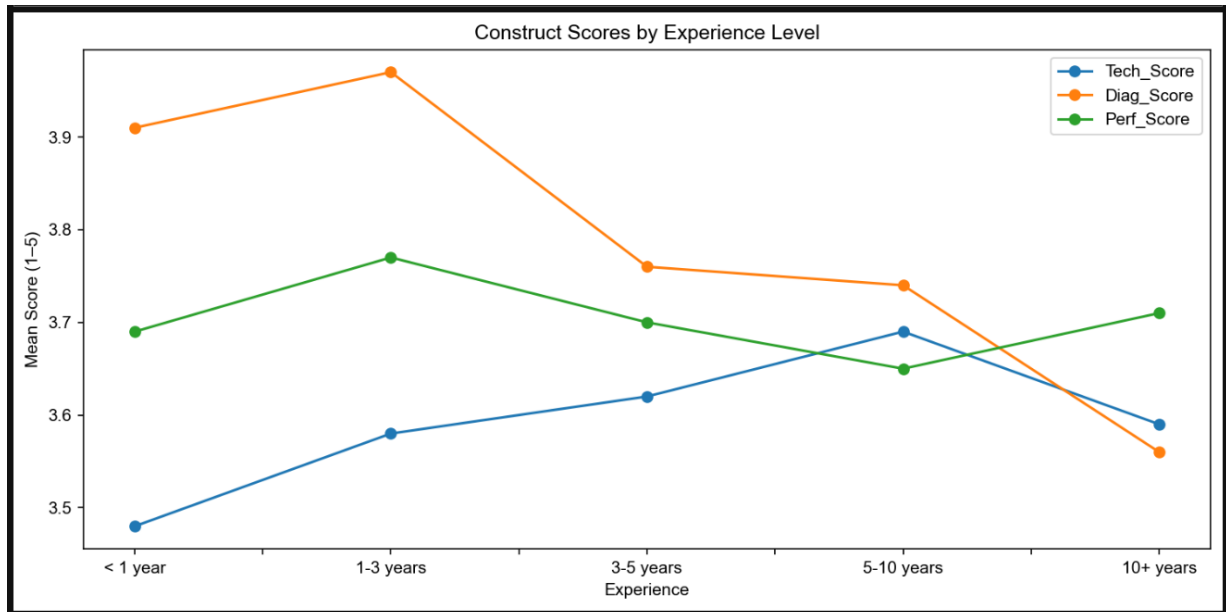
# Plot it
industry_analysis.plot(kind='bar', figsize=(12,5), colormap='Blues')
plt.title('Mean Construct Scores by Industry')
plt.ylabel('Mean Score (1-5)')
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.savefig('industry_crosstab.png', dpi=150)
plt.show()
```




```

exp_order = ['< 1 year', '1-3 years', '3-5 years', '5-10 years', '10+ years']
exp_analysis = df.groupby('Experience')[['Tech_Score', 'Diag_Score', 'Perf_Score']].mean()
exp_analysis = exp_analysis.reindex(exp_order).round(2)
exp_analysis.to_excel('experience_crosstab.xlsx')
exp_analysis.plot(kind='line', marker='o', figsize=(10,5))
plt.title('Construct Scores by Experience Level')
plt.ylabel('Mean Score (1-5)')
plt.tight_layout()
plt.savefig('experience_crosstab.png', dpi=150)
plt.show()

```



All analyses were conducted in Jupyter Notebook. The notebook was exported as PDF to provide a complete record of all code cells, outputs, and visualisations as proof of independent analytical work.

9% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **66 Not Cited or Quoted 7%**
Matches with neither in-text citation nor quotation marks
-  **15 Missing Quotations 2%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 4%  Internet sources
- 2%  Publications
- 7%  Submitted works (Student Papers)